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PROJECT TITLE: STUDENT PERFORMANCE AND STUDY-HOURS DASHBOARD

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STUDENT PERFORMANCE AND STUDY-HOURS DASHBOARD

Project Report

Python Programming Laboratory | Bachelor of Technology
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Abstract

This project presents a simple Python-based analysis of student assessment scores and reported study hours. The program reads anonymised student records from a CSV file, stores records in dictionaries, uses a StudentRecord class, calculates grades, applies NumPy for numerical statistics, uses Pandas for course-wise summaries, and generates scatter and histogram plots with Matplotlib. The included dataset contains 32 synthetic records across Python, Data Science and Algorithms. The analysis finds a strong positive association between study hours and scores in this sample; however, the dataset is illustrative and correlation does not establish causation.

Keywords: Python, CSV, student records, NumPy, Pandas, Matplotlib, correlation, data analysis

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1. Introduction

Student performance data often includes assessment scores and estimates of time spent studying. When such data is stored in a spreadsheet, it can be difficult to compare performance across courses, calculate summary statistics, or identify unusual values manually. Python provides a convenient way to automate these tasks and present results visually.

The Student Performance and Study-Hours Dashboard is a lightweight, beginner-friendly analytical project. It reads a CSV dataset, processes each student record, computes basic statistics, groups records by course, and visualises the relationship between reported study hours and assessment scores. The program is intentionally kept simple so that its logic can be understood and modified by students.

2. Problem Statement and Objectives

Problem statement: Investigate assessment scores and reported study hours using anonymised records. Implement score and grade functions, dictionary/CSV handling, a StudentRecord class, array-based statistics, course-wise aggregation, and plots. Evaluate whether an apparent association is influenced by outliers without making causal claims.

Objectives

• Read student information from a CSV file.
• Convert raw records into dictionaries and StudentRecord objects.
• Calculate assessment grades from scores.
• Calculate average, standard deviation and other summary measures using NumPy.
• Aggregate student performance by course using Pandas.
• Draw a scatter plot and score histogram using Matplotlib.
• Compare correlation before and after an outlier check.
• Explain the findings carefully and avoid causal conclusions.

3. Software and System Requirements

Component	Purpose
Python 3	Main programming language
CSV module	Read the dataset file
NumPy	Numerical arrays, mean, standard deviation and correlation
Pandas	DataFrame operations and course-wise grouping
Matplotlib	Scatter plot and histogram generation
Python IDLE / VS Code	Editing and running the program

The project can run on a desktop or laptop with Python 3 installed. NumPy, Pandas and Matplotlib are external packages and should be installed before execution.

4. System Design and Methodology

4.1 Dataset structure

The dataset is stored in **dataset.csv**. It contains anonymised identifiers and synthetic sample values. Each row represents one student record. The CSV has four columns:

Column	Description	Example
student_id	Anonymised student identifier	S001
course	Course name	Python
study_hours	Reported study hours per week	5
score	Assessment score out of 100	75

4.2 Processing workflow

1. Open the CSV file using csv.DictReader.
2. Read each row and create a StudentRecord object.
3. Convert objects into dictionaries for tabular analysis.
4. Create a Pandas DataFrame from the records.
5. Calculate averages and spread using NumPy.
6. Group the DataFrame by course and calculate mean values.
7. Compute the correlation between study hours and scores.
8. Check for outliers using the interquartile-range (IQR) rule.
9. Generate and save charts.

4.3 Grade calculation

The program uses the following simple grade mapping: A for scores of 90 or above, B for 80–89, C for 70–79, D for 60–69, and F for scores below 60. The grade is derived from the assessment score and is not stored as a required CSV field.

4.4 Statistical method

The mean represents the average value, while standard deviation describes the spread of values around the mean. Pearson's correlation coefficient ranges from −1 to +1 and measures the strength and direction of a linear association. The IQR is calculated as Q3 − Q1. A value is flagged as an outlier if it falls outside Q1 − 1.5×IQR or Q3 + 1.5×IQR. Outlier detection is used here as a sensitivity check, not as an automatic instruction to delete data.

5. Implementation of Major Modules

Module / function	Purpose
calculate_score()	Converts marks into percentage score
calculate_grade()	Assigns A, B, C, D or F
StudentRecord	Stores student ID, course, study hours and score
to_dict()	Converts a record into dictionary format
read_csv()	Loads records from dataset.csv
NumPy operations	Calculate average, standard deviation and correlation
Pandas groupby()	Summarise average study hours and scores by course
Matplotlib	Create scatter plot and score histogram

5.1 Object-oriented design

The StudentRecord class encapsulates the data for one student. Each object stores the student identifier, course, study hours and score. The grade() method calculates a grade, while to_dict() converts the object into a dictionary. This approach keeps individual records organised and makes the main analysis easier to read.

5.2 CSV handling

The standard Python csv module is used to read the file. DictReader maps each CSV row to a dictionary whose keys are the column headers. The code then creates StudentRecord instances from the values. This avoids the need for a separate database for a small demonstration project.

5.3 Data analysis and visualisation

NumPy is used for numerical calculations and correlation. Pandas is used to create a DataFrame and group records by course. Matplotlib produces a scatter plot of study hours against scores and a histogram showing the score distribution. The charts are saved as PNG images in the program directory.

6. Results and Discussion

The analysis below is calculated from the included 32-row synthetic CSV dataset. These values are demonstration results and do not represent real student outcomes.

Indicator	Observed value
Number of records	32
Mean assessment score	74.66 / 100
Median assessment score	76.00 / 100
Score standard deviation	15.86
Mean study hours	5.25 hours/week
Score Q1 / Q3	64.75 / 87.25
Score IQR	22.50
Outliers flagged (IQR rule)	0

6.1 Study hours versus assessment score

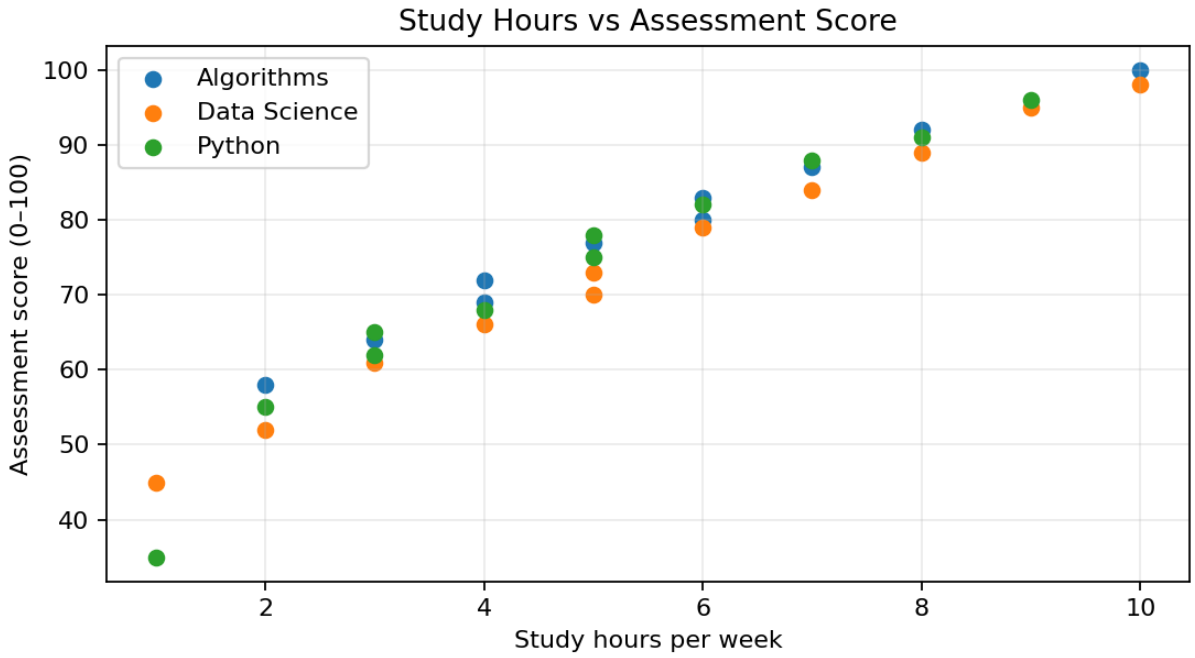


Figure 1. Scatter plot of reported study hours and assessment scores.

The Pearson correlation for this dataset is **0.972**. This indicates a strong positive linear association in the sample: higher study-hour values generally appear alongside higher scores. The association should not be interpreted as evidence that study hours alone cause higher marks.

6.2 Score distribution

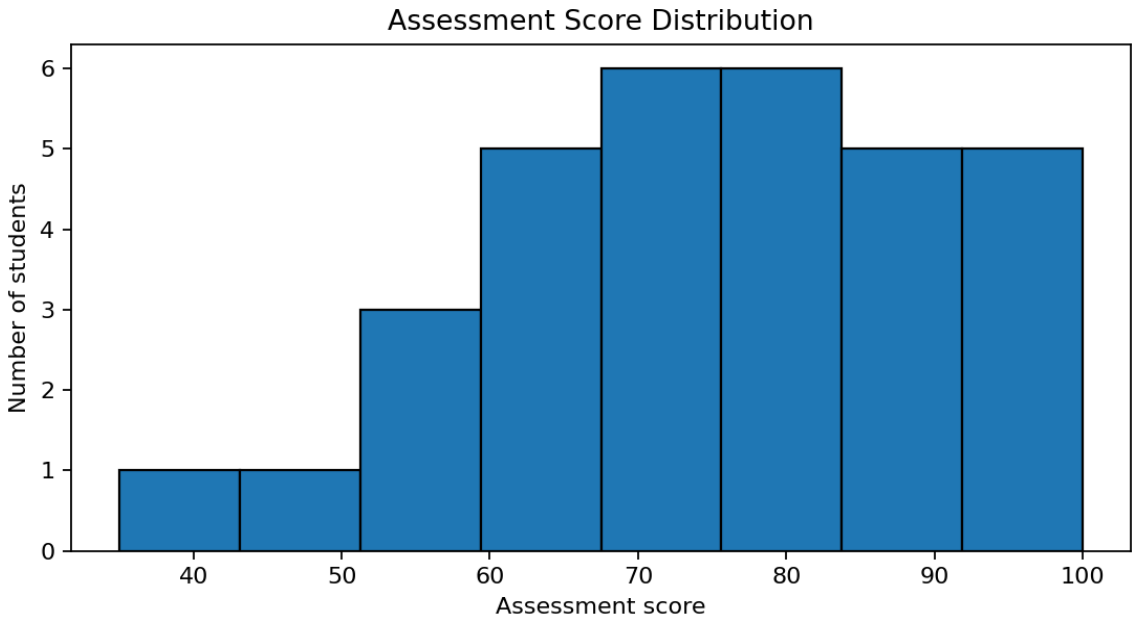


Figure 2. Histogram of assessment scores.

6.3 Course-wise summary

Course	Students	Mean study hours	Mean score
Algorithms	10	5.50	78.20
Data Science	11	5.45	73.82
Python	11	4.82	72.27

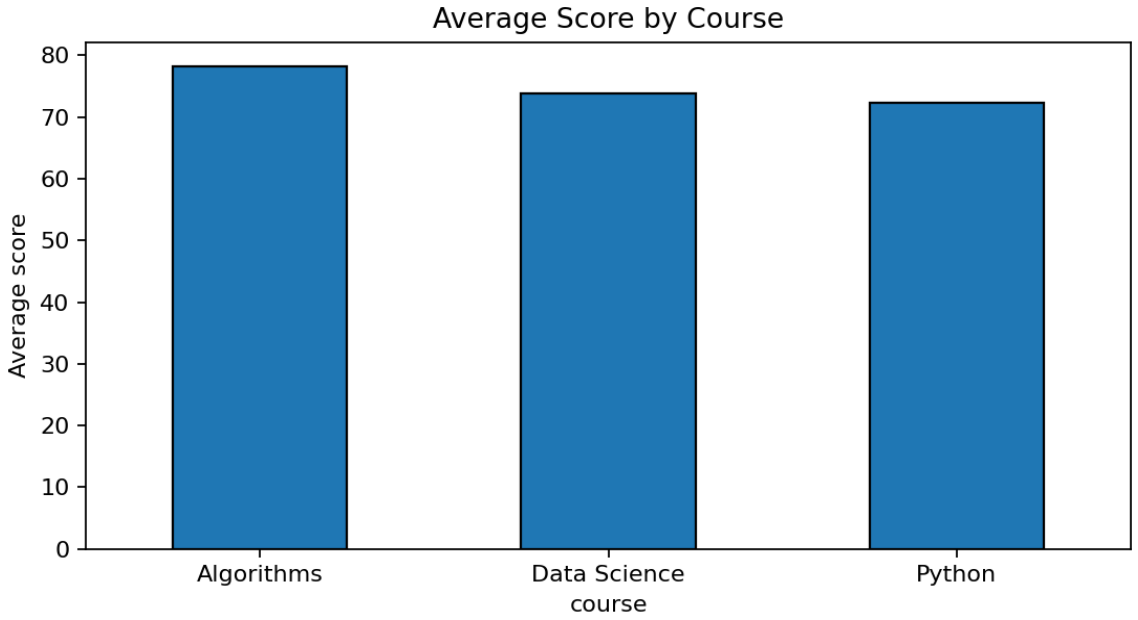


Figure 3. Average score by course.

Algorithms has the highest mean score among the three course groups in this dataset, followed by Data Science and Python. The differences are descriptive and should not be interpreted as course difficulty or teaching quality.

6.4 Outlier sensitivity

Using the 1.5×IQR rule, the score bounds are 31.00 to 121.00. The study-hour bounds are -3.00 to 13.00. The program flags 0 record(s) as outliers. The correlation before and after this check is 0.972; it is unchanged because no values in the dataset fall outside the IQR limits. This shows that the apparent association is not affected by the selected IQR rule for this particular CSV.

Important interpretation: Correlation does not imply causation. Factors such as prior knowledge, attendance, learning strategies, course content, motivation and measurement error could also influence assessment scores.

7. Testing and Validation

Test	Expected result	Status
Read CSV file	Rows are loaded from dataset.csv	Passed
Grade for score ≥ 90	Returns A	Passed
Grade for score 80–89	Returns B	Passed
Grade for score 70–79	Returns C	Passed
Grade for score 60–69	Returns D	Passed
Grade for score below 60	Returns F	Passed
Course aggregation	Mean values shown per course	Passed
Scatter plot	Chart saved to PNG	Passed
Histogram	Chart saved to PNG	Passed

The test cases correspond to the functions and workflow implemented in the simple Python project. Before submitting or presenting the project, the student should run the code locally and confirm that the required libraries are installed and the charts open correctly.

8. Limitations and Future Scope

Limitations

• The dataset is synthetic and does not represent actual students.
• Study hours are a simplified reported measure.
• The program is a basic analysis script rather than a full graphical dashboard.
• The grade thresholds are chosen for this demonstration and may differ from institutional rules.
• Correlation does not establish a cause-and-effect relationship.

Future improvements

• Add a Tkinter graphical user interface.
• Allow users to import different CSV files through a file picker.
• Add input validation and clear error messages for missing or invalid values.
• Add more statistical measures, such as median by course and Spearman correlation.
• Export a summary report automatically.
• Add semester-wise tracking and privacy protections for real student data.

9. Conclusion

This project demonstrates a simple end-to-end student performance analysis workflow using Python. It combines CSV file handling, dictionaries, an object-oriented StudentRecord class, NumPy calculations, Pandas aggregation and Matplotlib visualisation. The included synthetic dataset shows a positive relationship between reported study hours and assessment scores. The outlier check does not change the correlation for this dataset. The result is useful for learning data analysis, but it does not prove that studying more causes higher scores.

10. References

1. Python Software Foundation. Python 3 Documentation. https://docs.python.org/3/
2. NumPy Developers. NumPy Documentation. https://numpy.org/doc/

3. Pandas Development Team. pandas Documentation. https://pandas.pydata.org/docs/
4. Matplotlib Development Team. Matplotlib Documentation. https://matplotlib.org/stable/

Appendix: CSV Format and Execution

The dataset uses the following header format:

```
student_id,course,study_hours,score
S001,Python,2,55
S002,Python,3,62
S003,Python,4,68
```

Installation

```
pip install numpy pandas matplotlib
```

Execution

```
1. Keep main.py and dataset.csv in the same folder.
2. Open a terminal in that folder.
3. Run: python main.py
4. Read the statistical output in the terminal.
5. View the generated scatter plot and histogram.
```

Project files

main.py — Python program
dataset.csv — synthetic student dataset
README.txt — setup and run instructions

Team Details

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